

Generation Interconnection System Impact Study Report

For Queue Project Q21-1792

John R Kelly 138 kV

June 28, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1792. The Interconnection Customer proposes a 124.4 MW solar photovoltaic facility interconnecting at the John R Kelly 138 kV substation in the Virginia area. The request was received on June 22, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 631 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,034,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Morrell-Old Church 138.0 kV	Chancellor Row-Skaggs Island Jct 2 345.0 kV	129.1	232	5.3
Old Church-Tippecanoe 138.0 kV L	Skaggs Island Junction 2-Chancellor Row 345.0 kV	125.5	266	5.3
Skaggs Island Jct 2-Tippecanoe 138.0 kV	Skaggs Island Jct 2-Chancellor Row 345.0 kV L	123.1	295	5.3
Roney-Pana 138.0 kV Line	John R Kelly-Scott Paper Co 138.0 kV	120.8	235	10.2
Reeves Ave-Wheelabrator Portsmouth 138.0 kV	Mickens-Mira Vista 345.0 kV	119.7	245	5.1
Victory Way-Wheelabrator Portsmouth 138.0 kV Line	Wheelabrator Portsmouth-Gosport 138.0 kV	113.9	234	8.5
Hospital Distrito 7003-Hardy Reit 138.0 kV	Mitchell Dam-Skaggs Island Jct 2 Substation 345.0 kV	110.3	220	5.0
Elk Mound-Roney 138.0 kV	Scott Paper Co Substation-John R Kelly 138.0 kV L	107.8	222	10.2
Thole St-Reeves Ave 138.0 kV	Mickens-Mira Vista 345.0 kV	106.7	224	5.1
Harvard Substation-Hull St Road 138.0 kV	Skaggs Island Junction 2-Mitchell Dam 345.0 kV L	106.1	661	9.4
Bear Island 138/1kV TX	Chancellor Row-Bear Island 345.0 kV	105.8	513	7.2
Festa-Hospital Distrito 7003 138.0 kV	Mitchell Dam-Skaggs Island Junction 2 345.0 kV	105.8	197	5.0
Syl Laskin-Mitchell am 138.0 kV	Mitchell Dam-Haggin 345.0 kV	105.5	288	7.0
Hagin-Mitchell Dam 345.0 kV	Bear-Mecklenburg Power Station 345.0 kV	102.9	1,658	21.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fredericksburg-Possum Point 345.0 kV	Nauvoo-Birchwood 345.0 kV	102.5	1,600	6.9
Leavitt-Birchwood 345.0 kV	Bear Island-Dunn Sidng 345.0 kV	102.5	3,527	22.6
Nauvoo-Birchwood Substation 345.0 kV	Fredericksburg-Dunn Siding 345.0 kV	101.8	2,814	15.5
Chancellor Row-Natchez 138.0 kV	Mitcell Dam-Skaggs Island Jct 2 345.0 kV	100.4	599	6.2
Nauvoo Substation-City West Diesel Plant Substation 345.0 kV	Roberts Sw Sta-Nauvoo 345.0 kV Line	100.3	1,910	7.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Morrell Substation-Old Church 138.0 kV	Rebuild line with higher-capacity conductor	\$35,872,000	2% (\$844,000)
Tippecanoe-Old Church 138.0 kV	Rebuild line with higher-capacity conductor	\$5,331,000	2% (\$125,000)
Skagg Island Junction 2-Tippecanoe 138.0 kV	Rebuild line with higher-capacity conductor	\$5,319,000	2% (\$125,000)
Roney-Pana 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,514,000	46% (\$701,000)
Wheelabrator Portsmouth-Reeves Ave 138.0 kV	Rebuild line with higher-capacity conductor	\$4,743,000	44% (\$2,094,000)
Victory Way-Wheelabrator Portsmouth 138.0 kV Line	Rebuild line with higher-capacity conductor	\$7,343,000	7% (\$519,000)
Hardy Reit Substation-Hospital Distrito 7003 138.0 kV	Rebuild line with higher-capacity conductor	\$16,801,000	4% (\$685,000)
Elk Mound-Roney 138.0 kV	Reconductor line	\$636,000	46% (\$295,000)
Reeves Ave-Thole St 138.0 kV L	Reconductor line	\$1,835,000	44% (\$810,000)
Harvard Substation-Hull St Road 138.0 kV	Reconductor line	\$705,000	2% (\$12,000)
Bear Island 138/1kV TX	Replace transformer with higher-rated unit	\$5,073,000	1% (\$71,000)
Festa-Hospital Distrito 7003 138.0 kV	Reconductor line	\$6,870,000	4% (\$280,000)
Syl Laskin-Mitchell am 138.0 kV	Reconductor line	\$980,000	2% (\$16,000)
Hagin-Mitchell Dam 345.0 kV	Reconductor line	\$4,148,000	2% (\$68,000)
Fredericksburg-Possum Point 345.0 kV	Reconductor line	\$13,987,000	1% (\$199,000)
Leavitt-Birchwood 345.0 kV	Reconductor line	\$40,965,000	1% (\$584,000)
Nauvoo-Birchwood Substation 345.0 kV	Reconductor line	\$31,324,000	1% (\$439,000)
Chancellor Row-Natchez 138.0 kV	Reconductor line	\$453,000	1% (\$5,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Nauvoo Substation-City West Diesel Plant Substation 345.0 kV	Reconductor line	\$11,639,000	1% (\$162,000)
	Total allocated to project		\$8,034,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.